

A real phenotype and a real PGI on v9

40-minute lab, session 3

September 2026

Everything you touch today is real **Controlled Tier participant data**. Person-level files never leave the workspace. Printed outputs are aggregate, rounded, and suppression-screened by the scripts — keep them that way. There is no answer key: your numbers are real. This handout gives plausibility checks instead.

0–6 minutes: configuration and preflight

```
cd aou_cli_workshop
cp config.aou.example.sh config.aou.sh
nano config.aou.sh          # fill the values projected on screen
export WORKSHOP_CONFIG=config.aou.sh
bash scripts/00_preflight.sh
```

Fill `WORKSHOP_BUCKET`, `BILLING_PROJECT`, and `CDR_DATASET` from the values the instructor projects. `config.aou.sh` is git-ignored — it never gets committed. The `export` line matters in *every new terminal*: it is how the scripts know to run in AoU mode, where the disclosure gate blocks rather than warns.

6–12 minutes: stage genotypes, build the phenotype

```
bash scripts/10_fetch_aou_genotypes.sh
bash scripts/11_build_phenotype.sh
```

While the fetch runs (~1–3 minutes), read the two scripts. Then look at what arrived — two lines each, before trusting anything:

```
head -n 2 lab_data/chr22_filtered.fam    # FID convention?
head -n 2 lab_data/chr22_filtered.bim    # ID scheme?
head -n 2 lab_data/ancestry_preds.tsv    # key column name?
```

Checkpoints: samples in the fileset _____; variants _____.

The phenotype build queries height as program source concept **903133** — not the standard LOINC concept, which would silently pool in EHR heights — for a 1-in-10 teaching subset, then reduces to one row per person (median), applies 100–250 cm bounds, and standardizes within sex. Every one of those clauses is a decision; `scripts/build_aou_pheno.py` states each one.

12–18 minutes: phenotype EDA

```
bash scripts/12_pheno_eda.sh
cat results/aou_pheno_summary.txt
```

Open `results/aou_height_hist.png` and `aou_height_kde_sex.png` from the JupyterLab file browser. **Plausibility checks** (not answers): US-cohort sex-specific height means sit near 176 cm (male) and 162 cm (female) with SDs of roughly 7–8 cm; your subset should be in that neighborhood. Funnel counts print rounded to the nearest 100 — explain to your partner why.

18–28 minutes: score the PGI

```
bash scripts/13_score_real_pgi.sh
cat results/aou_pgi_summary.txt
```

Checkpoints: weight rows matched (variants processed) _____; skipped _____. A very low match rate means a wrong ID scheme or wrong column indices — stop and ask. Open `results/aou_pgi_kde_ancestry`.

This PGI uses one chromosome and teaching weights: it is deliberately partial. What matters is that you counted every match and every join on real data.

28–34 minutes: interpret with a partner

1. Three joins, three key spellings — name them.
2. Why the median per person? Name one alternative and its trade-off.
3. Say the two-sentence honest reading of the ancestry figure (distributions *and* accuracy).
4. What one variable changes to run everyone rather than 1-in-10? All 22 chromosomes?

34–40 minutes: save and finish

```
bash scripts/04_save.sh
```

In AoU mode the disclosure screen **blocks** the save on small-count findings; today's tables are designed to pass, and the screen's report will list your person-level files as workspace-only — read that list. Verify the cloud listing, **stop your cloud app** (VMs bill while idle), and finish the exit check: *I pulled real records, defended every cleaning step, counted every join, and read a real PGI honestly.*

Taking this to your own project

Change the concept ID, bounds, and one-row rule for your phenotype; swap in your trait's published weights; set `SAMPLE_MOD=1`; loop `CHROM` over 1–22 and sum the `.sscore SUM` columns. Rehearse in the sandbox first — everything except the BigQuery step tests locally for free.